

MUHAMMAD TALHA

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PROFESSIONAL SUMMARY

Final-year BS Cyber Security student at Air University with hands-on experience in penetration testing, digital forensics, network security, and secure system development. Completed cybersecurity internships in R&D and digital forensics and developed an AI-powered encrypted LoRa mesh communication system as a capstone project. Skilled in Kali Linux, Metasploit, Burp Suite, Wireshark, Nessus, Python, and secure protocol design. Certified in ISO/IEC 27001:2022, CAISR, and CSEDP. Seeking full-time opportunities in cybersecurity, SOC, information security, or security engineering.

EDUCATION

BS Cyber Security

2022 – June 2026 (Expected)

Air University, Islamabad

Relevant Coursework: Computer Networks, Network Security, Information Security, Information Assurance, Software Engineering, Mobile Device Security, Operating Systems, Data Structures & Algorithms, Digital Forensics, Reverse Engineering & Vulnerability Assessment

FSC Pre-Engineering

2019 – 2021

Fazaia Degree College Faisal, Karachi

Matric – Computer Science

2017 – 2019

Fazaia Degree College Faisal, Karachi

WORK EXPERIENCE

Cyber Security R&D Intern

Jan 2026 – Present

Tech Quadrant Pvt Ltd | Islamabad, Pakistan

- **Conducted** security research on RF and wireless network vulnerabilities, identifying attack vectors and security gaps across communication systems.
- **Performed** threat modeling, attack surface analysis, and risk assessment for embedded and wireless communication systems.
- **Supported** the design and testing of secure communication protocols, applying encryption and authenticated transmission techniques.

Digital Forensics Intern

Aug 2025 – Sep 2025

Cyborts | Remote

- **Executed** digital evidence acquisition and forensic disk imaging using industry-standard tools in accordance with forensic best practices.
- **Analyzed** disk images, system logs, and file system artifacts to identify indicators of compromise (IOCs) and suspicious activity patterns.
- **Produced** structured forensic investigation reports aligned with incident response procedures, suitable for legal and compliance review.

Accounts Manager

Aug 2024 – Dec 2025

TM Engineering Pvt Ltd | Rawalpindi, Pakistan

- **Managed** operational documentation, reporting, and compliance records using Microsoft Excel and internal business systems.
- **Prepared** monthly financial and audit reports using advanced Microsoft Excel, supporting executive decision-making and audit readiness.

PROJECTS

AEIS-LN: AI-Protected LoRa Mesh Communication System

Final Year Project

Technologies: ESP32-S3, LoRa, C++, Python, mbedTLS, AES, ECDH

- **Engineered** a fully decentralized, infrastructure-independent mesh network using custom AOMDV-inspired multi-path routing over LoRa (433 MHz) with dual ESP32-S3 nodes.
- **Implemented** ECDH (secp256r1) per-session key exchange, AES-128-CBC encryption, and HMAC-SHA256 Encrypt-then-MAC authentication via mbedTLS.
- **Designed** a three-stage AI firewall combining protocol compliance validation, metadata autoencoder anomaly detection, and DPI payload-level threat classification.
- **Built** a real-time web-based Mesh Node Console UI for live monitoring, packet transmission control, and firewall event logging.
- **Achieved** a false-positive rate below 3% through rigorous threshold optimization, validation, optimization, and technical documentation

Secure Remote Administration Simulation Tool

- **Developed** an encrypted command-and-control (C2) communication channel using Python socket programming with custom encryption layering.
- **Conducted** controlled penetration testing across isolated lab environments on both Kali Linux and Windows platforms.
- **Documented** security implications and recommended mitigation strategies for remote access vulnerabilities identified during testing.

Forensic Analyzer Application

Academic Project

- **Built** a digital forensic analysis tool capable of examining disk image (ISO) files and parsing system logs for forensic artifacts.
- **Performed** artifact extraction and structured log analysis to detect indicators of compromise (IOCs) and abnormal system behaviour.

CERTIFICATIONS & TRAINING

- [ISO/IEC 27001:2022 \(ISA\)](#) – Information Security Management - 2025
- [CCEP](#) – Certified Cybersecurity Expert Practitioner, Red Team Leaders - 2025
- [ICIP](#) – Introduction to Critical Infrastructure Protection, OPSWAT Academy - 2025
- [CAISR](#) – Certified AI Security Researcher, Red Team Leaders - 2026
- [CSEDP](#) – Certified Security Engineering & Development Professional, The SecOps Group - 2026
- [Certified AI-Data Security Specialist](#) - 2026

TECHNICAL SKILLS

Security Tools: Kali Linux, Metasploit, Burp Suite, Wireshark, Nessus, Snort (IDS), Autopsy, FTK Imager, Hydra, Binwalk, Scapy

Programming & Scripting: Python, SQL, C++

Security Domains: Penetration Testing, Vulnerability Assessment, Digital Forensics, Network Security, Incident Response, Threat Modeling, Secure Protocol Design

Other Tools: Microsoft Excel (Advanced), Git, Wireshark, VS Code

REFERENCES

Dr. Ijteba Ul Hasnain Shah

CEO, Tech Quadrant (Pvt) Ltd
info@tech-quadrant.com

Dr. Sheikh Qaisar Ayyub

Assistant Professor, Air University
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